Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/27/2016
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2012	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per NSSC information (original end date was 9/30/2016)--Ed., 4/23/13		
Key Personnel Changes/Previous PI:	none.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX12AR32G		
Performance Goal No.:			
Performance Goal Text:			
	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power		

Task Description:	compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably. NOTE: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status, effective October 2016, through continuation grant NNX16AT09G.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this EHD driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. For further technical and general information, please contact Professor Jamal Yagoobi at jyagoobi@wpi.edu . This device currently is not commercially available.
Task Progress:	The interactions between electric fields and flow fields were successfully studied in the presence and absence of liquid-vapor phase change and presence and absence of gravity, through extensive and varied experimental research. The contributions of this work are as follows: Operation of a meso-scale EHD conduction-driven single-phase flow heat transport system was confirmed in terrestrial and micro-gravity environments (for the first time). Single-phase data showed that the absence of gravity did not affect performance of EHD pumps based on conduction mechanism. For two-phase flow terrestrial experiments, the 1 mm diameter open-tube EHD pump was used as the sole driving mechanism for a meso-scale heat transport system. The system was able to transport 7.3 W of heat with just 0.015 W power input to the EHD pump. Successful recovery from evaporator dryout condition was also achieved for the first time in meso-scale for an EHD-driven two-phase flow heat transport system. A new and improved understanding of the relationship between net EHD pump pressure generation and downstream single- and two-phase flow pressure drop was also gained for this type of geometry and scale. The low mass fluxes generated by the EHD pump made it possible to measure the in-tube, local flow boiling heat transfer coefficients as a function of vapor quality along the evaporator length, providing limited but useful data points for this flow regime. In addition to the above results, the EHD conduction-driven two-phase flow heat transport system was tested successfully for 105 hours of intermittent operation without failure, where 5W of heat transport was consistently achieved with only 0.0042 W power input. EHD conduction pumping was successfully used for electro-wetting of a heater during liquid film flow boiling in terrestrial and micro-gravity conditions. A 62% increase in the critical heat flux was achieved in terrestrial experiments when the EHD conduction pump was activated. The experiments in micro-gravity were conducted for the first time on board a variable gravity parabolic flight. The transient results showed that the EHD conduction pump had an adverse effect on liquid film flow boiling below 5.0 W/cm2 due to delayed onset of nucleate boiling but eventually resulted in lowering of heater surface temperature at heat fluxes higher than 7.0 W/cm2. The most important determination, however, was that the EHD conduction-driven liquid film flow made the boiling process feasible in the absence of gravity. Experiments in liquid film flow boiling driven by EHD conduction pumping were also successfully completed for bare- and copper-plated nanofiber-enhanced surfaces. The results showed that the increased effective surface area and nucleation site density of enhanced surfaces led to lower surface temperatures. Like previous experiments, a delay in ONB (onset of nucleate boiling) and increase in surface temperature was observed during initial activation of the EHD pump at low heat fluxes. Beyond the isolated bubble regime, however, the combination of enhanced surface and EHD conduction pumping resulted in a maximum of 555% enhancement in boiling heat transfer coefficient compared to the bare surface experiment without EHD conduction pumping. The ratio of electric power consumption to heat removal rate was 1:55. The results were valuable as they provided new insights into EHD-driven liquid film flow and liquid-vapor phase change process in the presence and absence of gravity and paved the way for further experiments in micro-gravity. They also showed great promise for development of EHD conduction-driven heat transport systems for both terrestrial and micro-gravity thermal management applications. Finally, liquid film flow boiling and pool boiling in the presence of combined EHD conduction pumping and dielectrophoretic (DEP) force were successfully studied. The results provided new physical insights of these boiling processes in the presence and absence of electric fields. The combination of these two techniques resulted in up to 1217% enhancement in heat flux and heat transfer coefficient for a given superheat value with only 0.7% electric power input (compared to the corresponding heater power increase). A numerical simulation of the electric field for the DEP electrode/heater geometry used in the experiments was conducted. This was used to estimate the DEP force magnitude in comparison to the buoyancy force and applied to a basic heat transport model. The analysis was compared to experimental data to reveal an important understanding of the expected enhancement in the liquid film flow boiling curve. As with all other experimental results, the data were important for applications in thermal management in terrestrial and micro-gravity conditions. This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status, effective October 2016, through continuation grant NNX16AT09G. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 01/30/2026)

Articles in Peer-reviewed Journals	Patel VK, Seyed-Yagoobi J, Sinha-Ray S, Sinha-Ray S, Yarin A. "EHD conduction pumping driven liquid film flow boiling on bare- and nanofiber-enhanced surfaces." ASME Journal of Heat Transfer. 2016 Apr;138(4):041501-1-8. http://dx.doi.org/10.1115/1.4032021 , Apr-2016
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